

COVID-19 Patient Data Analysis Using Python

Minor Project Report

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Internship Duration: 8 weeks

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1. Introduction

The COVID-19 pandemic affected millions of people worldwide and generated a large amount of healthcare data. Analyzing this data helps identify trends, understand patient characteristics, and support data-driven decision-making.

This project performs exploratory data analysis (EDA) on a COVID-19 patient dataset using Python. Various visualizations are created to understand patient demographics and infection patterns.

2. Objectives

The objectives of this project are:

- Analyze COVID-19 patient records.
- Clean and preprocess the dataset.
- Perform exploratory data analysis.
- Visualize important trends using graphs.
- Extract meaningful insights from the data.

3. Dataset Description

The dataset contains information about COVID-19 patients, including:

- Patient Gender
- Birth Year
- Country
- State
- Region
- Infection Reason

The dataset was analyzed using Python libraries.

4. Tools and Technologies

- Python
- Jupyter Notebook

- Pandas
- NumPy
- Matplotlib
- GitHub

5. Methodology

The project was completed using the following steps:

1. Imported required Python libraries.
2. Loaded the patient dataset.
3. Explored the dataset.
4. Cleaned missing values.
5. Performed exploratory data analysis.
6. Created graphs.
7. Interpreted the results.
8. Documented the findings.

6. Data Cleaning

The following preprocessing tasks were performed:

- Checked missing values.
- Removed duplicate records if any.
- Handled null values.
- Verified data types.
- Prepared the dataset for analysis.

7. Exploratory Data Analysis

Several analyses were performed, including:

- Gender Distribution
- Birth Year Distribution

- Infection Reason Analysis
- State Distribution
- Country Distribution
- Region-wise Case Analysis

These analyses helped identify important trends within the dataset.

8. Visualizations

Figure 1: Gender Distribution

(Insert gender_distribution.png here)

Observation:

The graph shows the distribution of male and female patients. One gender has a higher number of recorded COVID-19 cases than the other.

Figure 2: Birth Year Distribution

(Insert birth_year_distribution.png here)

Observation:

Most patients belong to a particular birth-year range, indicating that certain age groups were more represented in the dataset.

Figure 3: Infection Reason Analysis

(Insert infection_reason.png here)

Observation:

The chart highlights the most common infection reasons. A few infection sources account for a large share of reported cases.

Figure 4: State Distribution

(Insert state_distribution.png here)

Observation:

The number of patients varies across different states, showing that some states recorded more cases than others.

Figure 5: Country Distribution

(Insert country_distribution.png here)

Observation:

Most records belong to one country, while only a few patients are from other countries.

Figure 6: Region-wise Distribution

(Insert region_distribution.png here)

Observation:

Some regions have significantly higher case counts than others, indicating an uneven geographical spread of COVID-19 cases.

9. Results and Observations

The analysis revealed several useful insights:

- The dataset contains both male and female patients.
- COVID-19 cases are concentrated in specific birth-year groups.
- Some infection reasons are more common than others.
- Certain states and regions report more patient records.
- The dataset mainly represents one country with a few records from others.

Overall, the visualizations provide a clear understanding of patient demographics and regional distribution.

10. Conclusion

This project successfully analyzed COVID-19 patient data using Python and data science techniques. The dataset was cleaned, explored, and visualized using multiple graphs. The findings demonstrate how exploratory data analysis can uncover meaningful patterns in healthcare data. This project also strengthened practical skills in data preprocessing, visualization, and analysis using Python.

11. Future Scope

The project can be extended by:

- Applying machine learning algorithms to predict recovery time.
- Building an interactive dashboard using Power BI or Tableau.
- Performing time-series analysis on COVID-19 cases.
- Developing predictive models for healthcare planning.

References

1. Python Documentation – <https://docs.python.org/>
2. Pandas Documentation – <https://pandas.pydata.org/>
3. Matplotlib Documentation – <https://matplotlib.org/>
4. COVID-19 Dataset used for this project